INTRODUCTORY REMARKS ON THE PROBLEM OF NOMENCLATURE IN RHEOLOGY.

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I.

The devising of an adequate nomenclature for the description of rheological phenomena is at once highly important and extremely difficult. It is important, because without an unequivocal language for the expression of results of experiments on deformable bodies, neither scientific discussion, nor well-planned technical application is possible. Its difficulty is due to the endlessly varying forms of behaviour presented by the manifold types of matter of which we constantly make use.

The Organising Committee of the Congress have thought that it would be appropriate to devote the introductory paper of the Proceedings to aspects of the problem of nomenclature. One might be tempted perhaps to make use of such an opportunity in order to lay down a set of definite rules, which every participant in our further activities would have to observe. We all know, however, that this is impossible at the present time. Nevertheless there is much to be said for giving some thought to these matters before starting with the more physical or chemical business of the congress, so as to prepare ourselves to recognize the existence of different modes of expression and the need of methods for translating various modes of expression into one another.

The Organising Committee was fortified in this opinion by the fact that the "Joint Committee on Rheology", instituted by the International Council of Scientific Unions, in its meeting in London of September 8-9, 1947, had decided to take up the subject of nomenclature, as a suitable theme for promoting international contact between rheologists. It is our hope that at the present Congress many scientists interested in this subject may meet one another and that their discussions may lead to a coordination of the work already undertaken in a number of countries, as well as to a widening of its scope. To further this aim a preliminary report has been prepared by Dr. Scott Blair and the present author, in which a number of concepts have been subjected to a critical examination. As this preliminary report is a Report to the "Joint Committee", it has been printed separately, and copies have been distributed to workers in this field. It is not included in the advance edition of the Proceedings, but an amended and extended version of it, in which account

is taken of amendments made at the Congress, will be published in th final edition.

The intrinsic difficulty of framing an adequate scheme of nomenclature faces us each time we attempt to give a detailed descrip ion of rheological phenomena and of the rheological characteristics o materials on which we are experimenting. In particular in every such description due regard must be paid to time effects, which by their multitudinous forms are continually presenting new complications. Agai and again one is driven to the conclusion that no single system of nomenclature seems able to embrace all phenomena.

In principle it may be argued that the properties of any sample non-living matter must be completely determined by its structure at the moment it is given into our hands, so that its behaviour in any subsequent series of experiments must be fully predictable when its present state is known. Hence it should be possible to base a proper physical system of nomenclature on an analysis of the structure of materials. Although this may be admitted in theory, the knowledge required for a description of structure sufficiently accurate to allo a prediction of rheological properties would have to penetrate very deeply into the interior of the sample, and it often looks as if data concerning the position and the state of motion of almost every atom present are necessary, nor must we forget to supplement them by a detailed description of the fields of force between the atoms.

In the majority of cases sufficient knowledge of this kind is un attainable with our present means of analysis. Many features remain hidden which nevertheless greatly influence the behaviour of a specim Consequently, in many cases we must substitute data on the history o the specimen for the unattainable data concerning its interior struct Actually we even make use of references to future behaviour: when we want to state that a certain percentage of a deformation is elastic, we refer to the behaviour shown by the specimen when the exterior fo acting on it are removed. Trivial as this remark may seem, neverthel such modes of expression sometimes make necessary an elaborate pres cription of experimental technique, since otherwise ambiguity may ar in the proper interpretation of the quantities we are measuring.

II.

In order to illustrate some of the possibilities and of the shortcomings of the physical description, let us briefly consider a few examples. We start from the simplest type of amorphous materials gases and ordinary liquids not containing molecules of extremely extended or extremely flattened shape, so that there are full potentialities for change of place and of orientation. We know that when a gas or an ordinary liquid is at rest and is left to itself, no subjected to exterior fields of force, the statistical distribution of the molecules in space and the distribution of their velocities obey definite rules, depending only on the temperature and the volum to which they are confined. With given values of these parameters th

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same physical state always appears, provided sufficient time is allowed for its establishment. Moreover the statistical properties of this equilibrium state are isotropic.

After any disturbance from the equilibrium state, however, some time of adjustment is needed before equilibrium is regained. Hence for short periods there may be present in any gas or liquid an anisotropy in the statistical distribution of the molecules and of their velocities. Such an anisotropy will lead to the appearance of stresses in the medium. When disturbances are repeated or changed periodically at intervals of time which are short compared with the so-called period of relaxation, the medium will show elastic behaviour, possibly with a relation between stress and strain of a nature similar to that assumed in the mathematical theory of an ideal elastic body. On the other hand, when the material is subjected to a continuous process of deformation, in particular when a stationary state of flow is set up, the resultant effect of deformation and relaxation leads to the appearance of a permanent anisotropic state of the molecular field, and thus to the appearance of a permanent system of stresses. As is well known this picture provides the basis for the theory of the viscosity of gases and liquids, a subject which is treated in its most modern form in the lecture to be given by Dr.Green. Without going into mathematical details it will be evident that the magnitude of the stresses increases both with the rate of deformation and with the duration of the time of relaxation, since increase of relaxation-time means a slower return to the equilibrium state and thus makes possible larger deviations from this state.

In the usual form of the theory of viscosity it is assumed that the rate of deformation is so slow compared with the relaxation process, that only slight deviations from the equilibrium state will be found. As the equilibrium state itself is isotropic, it follows that the directions characterising the anisotropy of the resulting field of molecular distribution are the same as the directions characterising the anisotropy of the field of flow - a fact which leads to parallelism and proportionality between the tensor of the viscous stresses and the tensor of the rate of deformation. This parallelism and proportionality form the basis of all hydrodynamical work on the flow of viscous liquids.

However, when the relaxation-time is large and the rate of deformation high, such a parallelism and proportionality will no longer be found, as the deviation from the equilibrium state will then show traces of a more distant past. Here we see manifested a kind of "memory" on the part of the flowing medium: as time is needed for readjustment, the structure at the present instant carries with it features of the past: when the memory is short, these features can be deduced from the instantaneous rate of deformation and we have the classical hydrodynamic theory; but when the memory extends over a longer period, more complicated relations appear.

Let us take the case of simple laminar flow, in which the velocity $\underline{u}$ is parallel to the $\underline{x}$-axis and is equal to $\underline{k}\,\underline{y}$ ($\underline{k}$ being a constant). The instantaneous change of position and form experienced by an element

of volume can be analysed into a translation of no interest, a rotation with angular velocity $\frac{1}{2}\underline{k}$ and a deformation process in which the element is stretched in a direction making angles of 45° with the positive $\underline{x}$- and $\underline{y}$-axes, while it is compressed in the perpendicular direction. Classical hydrodynamics calculates the stresses on this basis and resolves them into the well-known shearing stress acting parallel to the $\underline{x}$-axis in planes perpendicular to the $\underline{y}$-axis, combined with a shearing stress of no interest perpendicular and symmetrical to the former one. However, when the relaxation-time is considerable, the deformation process can no longer be described in this way, and the stresses must be calculated from the deformation experienced by an element of volume in a period of finite duration, i.e. they must be calculated from finite displacements parallel to the $\underline{x}$-axis. It is then found that the principal directions of extension and compression are no longer in the 45° directions, but are rotated through a certain angle in such a way that the direction of extension comes nearer to the $\underline{x}$-axis. This has effects on the stress system and can be interpreted by saying that to the viscous stresses of classical hydrodynamics there appears to be added a tension along the $\underline{x}$-axis, i.e. in the direction of the streamlines in this case.

Situations of this kind, in which there is a difference of orientation between the stress tensor and the tensor of instantaneous deformation, due to the fact that the stress tensor is influenced by features of a more remote past, lead to relations in which the stress tensor must be expressed as a non-linear function of the tensor of the instantaneous rate. Relations of this kind are considered in the lecture to be given by Dr.Weissenberg, although the mode of description chosen by him is more abstract and general than the picture given here.

This reasoning illustrates one of the ways in which a non-linear relation between rate of deformation and stress can arise. There are other possibilities; e.g. the relaxation phenomena may depend on the magnitude of the deformations produced. I will ignore such cases, but would point to another feature of importance. We have spoken of the relaxation time as if this were always a single well defined quantity. This assumption will probably be valid for gases and ordinary liquids with only one species of molecules, the shapes of which do not differ too much from a sphere. When the molecules are much more extended in one direction than in another and also when there are present several types of molecule, relaxation phenomena may require more than one time constant for their specification. The relation between stress and strain then will become still more complicated, and it will be evident that deviations may again occur between the directions of the tensors of stress and of instantaneous rate of strain. In Dr.Gray's lecture before section a a reference is made to systems consisting of components with molecules of greatly differing shapes.

In many materials we must reckon with a complicated "spectrum" of relaxation-times. Some members of this spectrum may be very large, even practically infinite. What will be the effect of the presence of infinite relaxation-times? In such a case it will not be possible to

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reach a steady state of flow in finite time with a constant field of stress. In other words ordinary liquid flow will not be found. With a constant shearing stress the rate of shear will either continually decrease or perhaps increase the latter (if large deformations should lead to a reduction of the forces between the molecules). Materials for which this is the case behave as "fluids", provided there are still also finite relaxation times in the spectrum.

If - for small deformations at least (i.e. when small stresses are applied) - there are no, or practically no, finite relaxation-times, so that no relaxation is observable, the material is said to behave as a solid. It will be a plastic solid (or plastic substance) if, for deformations surpassing a certain limit, relaxation sets in, thus again making flow possible.

Thus far, structural considerations seem to provide a valuable basis for a system of nomenclature. But there is still a further point to be considered. We have started from the idea of a medium which may be considered as on the average homogeneous. There are, however, always deviations from the ideal mean state, which deviations will themselves occur according to statistical laws so long as we are concerned with equilibrium. In many cases these deviations may be of slight importance. We must, however, reckon with the possibility that when the medium is in a state of flow, the deviations may increase, even when originally of atomic dimensions or referring to rows or parts of planes of atoms. When relaxation operates quickly, the ultimate dimensions reached will remain small. However, when relaxation is slow, it is quite possible that deviations will reach submicroscopic dimensions. In that case the internal structure of the material may acquire a pattern of irregularities of great importance to the field of stresses.

I do not know whether in the case of the flow of liquids or fluids there exist clear indications of the presence of such a process. But we are immediately confronted with it when we pass to plastic materials. We all know that the subject of permanent deformation of solid bodies is rendered extremely difficult by the structural changes produced in the materials - structural changes which must be largely due to the increase and the redistribution of inhomogeneities (irregularities of molecular pattern, faults in atomic arrangements, dislocations, crystal boundaries, cracks etc.).

These considerations show that, although a description and a nomenclature based on structural considerations may at first sight appear relatively simple and straight forward, increasing complications are met with when we come to cases where infinite relaxation-times must be taken into account, especially for solids and plastic materials. A process of deformation then can store up in the material so many deviations from the normal (or ideal) structure, that it becomes impossible to describe them so as to explain adequately the actual rheological behaviour of the material.

III.

Leaving now the idea of a description based on structure, we turn to the phenomenological approach, that is, to a description of rheological behaviour under specified circumstances.

Working within the framework of Newtonian mechanics, it seems appropriate to start our specifications from the kinematics of deformation and from an analysis of forces and stress systems, as devised originally for the description of the behaviour of homogeneous elastic bodies and of simple viscous liquids. When due regard is given to time as an independent coordinate, a descriptive system can be found, which provides a systematic picture of a number of phenomena and which proves to be of great value as a basis for the rigorous analysis of the relations between deformation, stress and time. This is the mechanical picture developed in a report on the nomenclature of deformations, prepared during the war years by the Committee for the Study of Viscosity and Plasticity of the Royal Netherlands Academy of Sciences, which has served as a model for the preparation of the preliminary report by Dr.Scott Blair and myself already referred to.

I think it superfluous to give a summary of its principles here, but I will point out that in this system also we perceive limitations. There are many cases where the rigorous application of the principles adopted, although logically fully justified, leads to descriptions which may seem clumsy to practical workers in the field.

This is apparent, for instance, when part of a deformation is recoverable (elastic) and part non-recoverable (permanent). A proper description according to the scheme prescribed by mechanics requires that during the execution of a deformation process, both parts should be known separately at every instant. However, we cannot simply see which part is recoverable and which not, because we cannot penetrate into the internal structure. Consequently a measurement of the elastic part can only be performed by unloading the specimen under observation. This means that we must interrupt the test. As most substances suffer internal changes in consequence of which an experiment cannot be repeated on the same specimen, the course of the elastic deformation as a function of the time can be properly measured therefore only by performing experiments on a series of specimens, each of which is subjected to a deformation process of limited duration, which durations must increase with suitably chosen steps.

In a similar way the correct determination of the relation between rate of shear and shearing stress requires the performance of a series of experiments, in each of which a definite exactly constant rate of shear is set up. The correct determination of hardening or softening curves requires still more circumspection.

When one looks at all the rules which must be observed, the problem arises whether the mechanical scheme of description is suitable for every possible case, or whether it sometimes seems to be moving too far away from practical points of view and may become too artificial. This

problem assumes an even more poignant character when one thinks of the complicated cases of rheological behaviour which are not covered or only partially covered by the preliminary report, and which are referred to by such terms as "large elastic deformation", "Spinnbarkeit", "flow-elasticity", "climbing phenomena", "tack", "thixotropy", "false-body", "rheopexy" and "structural viscosity". Problems of fracture also should be considered. (It is hoped that a number of these subjects may be discussed at the Congress, and it is intended to give some attention to them in the final edition of the report on nomenclature.)

I believe nevertheless that the "mechanical" description must give even in such cases the more fundamental picture. But it is understandable that in many methods of testing approaches other than mechanical are followed. For instance very often the total deformation will be recorded as a function of the time without interrupting the process; only at the end of the experiment are all exterior forces removed from the specimen and the recovery is investigated. The result of such an experiment cannot be analysed in the rigorous mechanical way which we consider necessary for a complete description. Nevertheless, the knowledge of the total deformation as a function of the time in an uninterrupted experiment is of great value and can give data of importance in relation to the use to be made of the material tested. Many more of such examples can be given: indeed most actual testing practice presents them.

It would be a great error to banish such methods; we must admit that from a general point of view there are many ways of analysing rheological phenomena. We should not forget that in actual practice (and rheology is an eminently practical subject) rheological phenomena present themselves as wholes, as complex processes, in which it is often not even our aim to retain a definite physical state of the material, but in which temperature, degree of polymerisation, chemical constitution, pattern of crystallisation etc. all may change - these changes not occurring at random, but being brought about for a definite technological purpose. Processes of moulding are good example, and I presume that Dr.Houwink's paper on Rheology in Industry will give further illustrations of this.

In view of this complexity we must recognize that there are various legitimate methods of description, Having regard to a very useful analogy, we may say that there must be various ways of projecting the infinite variety of possible rheological phenomena on to suitable frames of reference - just as a multidimensional geometrical pattern can be projected on to various systems of coordinates; some restricted to a few coordinates only, others embracing projections on many axes. We may prefer one of these systems of coordinates as being more in tune with classical mechanical reasoning than another, but this does not do away with the fact that special systems have their interest for particular purposes. For instance certain of the principles of classification proposed by Sir Charles Goodeve, at least in some of his former publications, are based on a combination of properties which mechanical analysis separated. This may well be interpreted by saying that Goodeve applies a different method of projection.

Now it is one of the aims of science - we have learned this particularly in physics - to find transformation formulae, which enable us to pass from one system of coordinates to another. In the domain of rheology this means that we must find rules which will enable us to pass from one system of analysis of rheological phenomena to another. The situation is of course much more complicated than is the case with the projections of a geometrical figure on a system of coordinates. Nevertheless, geometry provides us with a characteristic which plays a large part in all applications of the idea of projection. We all know the importance of rectangular coordinates; they allow us to obtain independent data concerning the extension of our geometrical system in different directions, and when projections have been obtained on the axes of one particular rectangular system, the calculation of the projections on the axes of a differently orientated system is easy.

Hence in devising methods for the description of rheological phenomena, we must strive to analyse these phenomena in such a way that the various data obtained are independent of each other. This aim is realised by the mechanical description used in the preliminary report, and this is its particular merit. On the other hand an analysis, based on tests where a specimen is subjected to a number of deformation processes in which, for example, only the total deformation is observed (perhaps for different rates of deformation or with harmonically oscillating deformations of various frequencies and amplitudes) will not give independent data, although the data obtained may be important from a practical point of view.

This is even more true when various industrial methods of testing are considered.

By means of suitable ways of comparing the results of different methods of testing, it is now possible to find out in how far these results are interdependent; this is done by determining the degree of correlation between various sets of results. This idea can be applied even to those methods of testing where subjective data are used, as for instance, the feeling of stiffness obtained by putting one's finger into the material, or other methods of "handling", of which Mr. Harper's paper gives an example.

The paper to be read by Dr. Scott Blair on Psycho-physical aspects of Rheology gives a general discussion of this problem. Scott Blair points out how it is possible with the aid of "factorial analysis" to determine the degrees of interdependence between the results of various tests and in this way to establish "normal coordinates", a procedure which leads to very interesting conclusions.

I am sure that our subject can be considerably developed in this direction. It may be too arduous a task to deduce transformation formulae from every system of description and corresponding nomenclature to every other, but I feel certain that much can still be done. For my part, I would always consider the mechanical "Newtonian" description advocated in our preliminary report as something like the "Euclidean" description of elementary geometry. Modern developments in physics have taught us

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to criticise the old-fashioned view that it has unique validity for the physical universe, but it still remains a useful scheme of comparison. I am convinced that the mechanical description gives the best basis for establishing a sound link with the analysis of physical structure.

These considerations must be kept in mind when we attempt to devise schemes for the nomenclature of rheological phenomena. It will be necessary to make a kind of dictionary with many entries and a reference system must be developed which shows the interconnections. Our most useful work will be the attainment of extreme clarity in the exposition of these interconnections.

One final remark may be made. Although probably everybody present will admit the importance of uniformity in nomenclature, it will be evident that no such uniformity was to be found in the manuscripts presented to the Organising Committee. The editors of these Proceedings have not deemed it fit to introduce on their own accord such changes as would have been necessary to obtain uniformity.

Indulgence therefore is asked from the reader of these papers, when he finds that the rules advocated have not always been observed.